Series and Integral Transforms

Class Notes Summary — Weeks 1–3

Sources: lecture notes + solved exercises from class sessions

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Key: [**ILLEGIBLE: description**] = handwritten content that could not be read; filled by mathematical context where possible.

Contents

1 Positive Series — Convergence Tests, Cont. (18.3)

1.1 Integral Test and Error Estimation (typed content)

From the integral test, the sum of the generalized harmonic series is bounded:

$$\int_1^\infty f(x)\,dx \;\le\; \sum_{n=1}^\infty a_n \;\le\; a_1 + \int_1^\infty f(x)\,dx.$$

Case $p = 2$, $f(x) = 1/x^2$: We obtain $1 \le \sum 1/n^2 \le 2$. Using the partial sum $S_3 = 1 + \frac{1}{4} + \frac{1}{9} \approx 1.3611$ and the tail integral:

$$S_3 + \frac{1}{4} \;\le\; \sum_{n=1}^\infty \frac{1}{n^2} \;\le\; S_3 + a_4 + \frac{1}{4}, \qquad \text{giving } 1.6111 \le \sum \frac{1}{n^2} \le 1.6736.$$

(True value: $\pi^2/6 \approx 1.6449$.)

Example 1.1 ($\sum \frac{1}{n \ln n}$ diverges — typed)**.** $f(x) = 1/(x \ln x)$, substitution $u = \ln x$, $du = dx/x$:

$$\int \frac{dx}{x \ln x} = \int \frac{du}{u} = \ln(\ln x) \to \infty.$$

So the series diverges.

Example 1.2 ($\sum_{n=2}^\infty \frac{1}{n \ln n \ln(\ln n) \cdots}$)**.** Repeated substitution gives divergence for any finite chain of iterated logarithms. (General principle: [**ILLEGIBLE: precise statement of iterated log divergence criteria**].)

1.2 Ratio and Root Tests (18.3 — mixed typed/handwritten)

Definition 1.3.

$$L = \limsup_{n\to\infty} \sqrt[n]{a_n} \quad \text{(root test / Cauchy)}, \qquad L = \lim_{n\to\infty} \frac{a_{n+1}}{a_n} \quad \text{(ratio test / D'Alembert)}.$$

Theorem 1.4. *If $L < 1$: converges. If $L > 1$: diverges. ($0 \le L \le \infty$.)*

Key relation (typed): If the ratio limit exists, then the root limit also exists and equals the same value. The converse is false — root test is strictly stronger.

Proof sketch for convergence ($L < 1$) — typed: Set $q = (L+1)/2 < 1$. There exists N such that for all $n > N$, $a_{n+1}/a_n < q$. Then $a_{N+k} < q^k \cdot a_N$, so $\sum a_n$ is bounded by a geometric series. □

Proof sketch for divergence ($L > 1$) — typed: There exists N such that $a_{n+1}/a_n > 1$ for all $n > N$, so $a_n \not\to 0$. □

Example 1.5 (Interleaved sequence — handwritten, partially legible)**.**

$$\{a_n\} = \left\{1, 1, \tfrac{1}{2}, \tfrac{1}{3}, \tfrac{1}{4}, \tfrac{1}{9}, \tfrac{1}{8}, \tfrac{1}{27}, \ldots\right\}, \quad a_n = \begin{cases} 1/2^{(n-1)/2} & n \text{ odd} \\ 1/3^{(n-2)/2} & n \text{ even.} \end{cases}$$

Ratio test for odd n: $a_{n+1}/a_n = \frac{1/3^{(n-1)/2}}{1/2^{(n-1)/2}} = (2/3)^{(n-1)/2} \cdot$ [**ILLEGIBLE: factor**] $\to 0$.

Ratio test for even n: $a_{n+1}/a_n = \frac{1/2^{n/2}}{1/3^{(n-2)/2}} = (3/2)^{(n-2)/2} \cdot$ [**ILLEGIBLE: factor**] $\to \infty$. So the ratio oscillates; generalized ratio test fails (even as a positive series!).

Root test:

$$\sqrt[n]{a_n} = \begin{cases} 2^{-(n-1)/(2n)} \to 1/\sqrt{2} & n \text{ odd} \\ 3^{-(n-2)/(2n)} \to 1/\sqrt{3} & n \text{ even.} \end{cases}$$

Both limits are < 1, so $\limsup \sqrt[n]{a_n} = 1/\sqrt{2} < 1$. Converges.

2 Positive Series — Completion (22.3)

Example 2.1 ($\sum x^n/n^2$ — handwritten)**.** Ratio test: $a_{n+1}/a_n = x \cdot n^2/(n+1)^2 \to x$.

- $|x| < 1$: converges.
- $|x| > 1$: diverges.
- $x = 1$: $\sum 1/n^2$ converges (p-series).
- $x = -1$: [**ILLEGIBLE: check at x=-1; likely converges conditionally**].

Root test gives the same: $\sqrt[n]{x^n/n^2} = x \cdot n^{-2/n} \to x$.

2.1 Cauchy Sequences (22.3 — typed)

Definition 2.2 (Cauchy Sequence)**.** $\{a_n\}$ is Cauchy if $\forall \varepsilon > 0\ \exists N : \forall n, m > N,\ |a_n - a_m| < \varepsilon$.

Theorem 2.3 (Cauchy's Criterion)**.** *A sequence converges $\iff$ it is Cauchy.*

Proof sketch (typed):

- $(\Rightarrow)$: $|a_n - a_m| \le |a_n - L| + |L - a_m| < 2\varepsilon$.
- $(\Leftarrow)$: Cauchy $\Rightarrow$ bounded $\Rightarrow$ convergent subsequence (Bolzano–Weierstrass) $\Rightarrow$ full sequence converges.

Why define this? (typed)

1. Convergence detected without knowing the limit.
2. Generalizes to metric spaces.
3. Defines $\mathbb{R}$ as equivalence classes of Cauchy sequences of rationals.

3 Absolute Convergence (25.3)

Setup (typed): The only situation where positive-series tests cannot directly apply is when a series has infinitely many positive *and* infinitely many negative terms.

Definition 3.1.

- $\sum a_n$ is **absolutely convergent** if $\sum |a_n| < \infty$.
- $\sum a_n$ is **conditionally convergent** if it converges but $\sum |a_n| = \infty$.

Theorem 3.2. *Absolutely convergent $\Rightarrow$ convergent.*

Proof sketch (typed): $\sum |a_n|$ converges $\Rightarrow$ its partial sums are Cauchy $\Rightarrow$ partial sums of $\sum a_n$ are Cauchy (triangle inequality).

Example 3.3 (Typed)**.** $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!}$: the absolute-value series is $\sum \frac{1}{(2n)!} = \cosh(1) \approx 1.543$. By ratio test: $|a_{n+1}/a_n| = 1/((2n+1)(2n+2)) \to 0$. Converges absolutely. Sum $= \cos(1) \approx 0.540$.

Example 3.4 (Ratio/Root for general series — partially handwritten)**.** Consider $a_n = \begin{cases} 5/7^n & n \text{ odd} \\ -1/7^n & n \text{ even} \end{cases}$, i.e. $\sum \frac{2+3(-1)^n}{7^n}$.

Ratio: $|a_{n+1}/a_n|$ alternates between $|-1/5|$ and $|-5|$ — oscillates, inconclusive.

Root: $\sqrt[n]{|a_n|}$. For odd n: $(5/7^n)^{1/n} = 5^{1/n}/7 \to 1/7$. For even n: $(1/7^n)^{1/n} = 1/7$. So $\limsup \sqrt[n]{|a_n|} = 1/7 < 1$. Converges absolutely.

Example 3.5 ($\sum x^n/n^2$ (absolute convergence domain) — handwritten)**.** $|a_{n+1}/a_n| = |x| \cdot n^2/(n+1)^2 \to |x|$. Absolutely convergent for $|x| < 1$. At $|x| = 1$: $\sum 1/n^2$ converges, so absolutely convergent on the whole boundary too. For $|x| > 1$: diverges (terms $\not\to 0$).

4 Alternating Series (29.3)

4.1 Leibniz's Theorem (typed)

A series $\sum a_n$ is called **alternating** if $a_n = (-1)^n b_n$ with $b_n \geq 0$.

Theorem 4.1 (Leibniz)**.** *If $b_n \geq 0$ and $b_n \searrow 0$ monotonically, then $\sum_{n=1}^{\infty}(-1)^{n+1} b_n$ converges. Its sum S satisfies $S_{2n} \leq S \leq S_{2n+1}$ and $|S - S_n| \leq b_{n+1}$.*

Proof (handwritten, reconstructed). Even partial sums:

$$S_{2n} = (b_1 - b_2) + (b_3 - b_4) + \cdots + (b_{2n-1} - b_{2n}) \geq 0.$$

Each bracket is ≥ 0 (since b_n decreasing), so $\{S_{2n}\}$ is increasing. Also $S_{2n} = b_1 - (b_2 - b_3) - \cdots - (b_{2n-2} - b_{2n-1}) - b_{2n} \leq b_1$, so bounded.

Odd partial sums:

$$S_{2n+1} = b_1 - (b_2 - b_3) - (b_4 - b_5) - \cdots - (b_{2n} - b_{2n+1}) \leq b_1.$$

$\{S_{2n+1}\}$ is decreasing. Since $S_{2n+1} - S_{2n} = b_{2n+1} \to 0$, both subsequences converge to the same limit S. Therefore $\{S_n\}$ converges. □

Remark. A Leibniz series cannot distinguish absolute from conditional convergence on its own.

Example 4.2 (Alternating harmonic series (typed))**.** $\sum_{n=1}^{\infty}(-1)^{n+1}/n = 1 - 1/2 + 1/3 - 1/4 + \cdots = \ln 2 \approx 0.693$.

Partial sums: $S_1 = 1$, $S_2 = 1/2$, $S_3 = 5/6$, $S_4 = 7/12 \approx 0.583$, $S_5 \approx 0.783$, $S_6 \approx 0.617$.

Numerical acceleration (typed):

$$\frac{S_4 + S_5}{2} \approx 0.683, \qquad \frac{S_5 + S_6}{2} \approx 0.700.$$

Exact: $S = \ln 2 \approx 0.693$.

Example 4.3 ($\sum(-1)^n/n!$ (handwritten)). $\sum 1/n! = e$, so $\sum(-1)^n/n!$ converges absolutely. Sum $= e^{-1} \approx 0.368$. Partial sums $S_0 = 1, S_1 = 0, S_2 = 1/2, S_3 = 1/3, S_4 \approx 0.375$, $S_5 \approx 0.367$, $S_6 \approx 0.368$. Bounds: $0.367 \le S \le 0.368$. ✓

Example 4.4 ($\sum(-1)^{n+1}(n-2)^2/n^3$ — handwritten). $b_n = (n-2)^2/n^3$. Check $b_n \to 0$: yes. Check decreasing: $b_n - b_{n+1} = \frac{n}{2n-1} - \frac{n+1}{2n+1} = \frac{1}{4n^2-1} > 0$. Yes. Converges by Leibniz. Not absolute: $b_n \sim 1/n$, compare $\sum 1/n$ diverges.

Example 4.5 ($\sum(-1)^{n+1}\frac{n}{2n-1}$ — typed). $b_n = n/(2n-1) \to 1/2 \neq 0$. Necessary condition fails. Diverges.

4.2 Algebraic Properties of Series (typed)

1. $\sum ca_n = c\sum a_n$ and $c\sum a_n + c\sum b_n = c\sum(a_n + b_n)$.
2. **Grouping**: inserting parentheses (same order) gives a grouping.
   If series converges $\Rightarrow$ every grouping converges to same sum.
   Converse fails (e.g. $\sum(-1)^n$).
   Exception: if all terms in each group have the same sign, grouping converging $\Rightarrow$ original converges.
3. Changing finitely many terms does not affect convergence.

Example 4.6 (Grouping does not imply convergence — typed). $\sum_{n=0}^{\infty}(-1)^n$: grouping gives $(1-1)+(1-1)+\cdots = 0$, but partial sums $\{1,0,1,0,\ldots\}$ diverge.

4.3 Rearrangements (typed)

Theorem 4.7 (Cauchy). *Rearranging terms of an* absolutely *convergent series preserves the sum.*

Theorem 4.8 (Riemann Rearrangement). *If $\sum a_n$ converges conditionally, for any $L \in \mathbb{R} \cup \{\pm\infty\}$ there is a rearrangement with sum L.*

Sketch (typed). Define $p_n = \max(a_n, 0)$ and $N_n = \min(a_n, 0)$. Since the series is conditionally convergent:

$$\sum |a_n| = \sum(p_n + N_n) = \infty, \quad \sum p_n = +\infty, \quad \sum N_n = -\infty.$$

To reach L: greedily take positives until exceeding L, then negatives until below L; repeat. Overshoot $\to 0$ since $a_n \to 0$.
To diverge to $+\infty$: take enough positives to exceed 1, one negative, enough positives to exceed 2, etc.
"Wild" divergence: alternate exceeding $+1$ then -1 **[ILLEGIBLE: precise block construction]**. □

Example 4.9 (Rearranging to $2\ln 2$ — handwritten, partially legible).

$$\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\cdot 2 = 1 + \frac{1}{2} - \frac{1}{2} + \frac{1}{3} + \frac{1}{4} - \frac{1}{4} + \cdots$$

[ILLEGIBLE: exact grouping/block argument for $2\ln 2$; blocked partial sums approach shown but not fully legible]

5 Solved Exercises from Tutorial Sessions

5.1 SIT1 — Tutorial Session Continuation (18.3)

The following are exercises from Sheet 1 solved in the 18.3 tutorial session.

Exercise 1.5(): $\sum_{n=1}^{\infty} \frac{1}{(2n+1)!}$

Solution: Ratio test:

$$\frac{a_{n+1}}{a_n} = \frac{(2n+1)!}{(2n+3)!} = \frac{1}{(2n+2)(2n+3)} \to 0 < 1.$$

Converges.

Exercise 1.5(): $\sum_{n=1}^{\infty} \frac{2n-1}{3^n}$

Solution: Root test:

$$\sqrt[n]{\frac{2n-1}{3^n}} = \frac{\sqrt[n]{2n-1}}{3} \to \frac{1}{3} < 1.$$

Converges.

Exercise 1.5(): $\sum_{n=1}^{\infty} \frac{1}{2^n} \left(1 + \frac{1}{n}\right)^{n^2}$

Solution: Root test:

$$\sqrt[n]{\frac{1}{2^n} \left(1 + \frac{1}{n}\right)^{n^2}} = \frac{1}{2} \cdot \left(1 + \frac{1}{n}\right)^{n} \to \frac{e}{2} > 1.$$

Diverges.

Exercise 1.6(): $\sum_{n=0}^{\infty} \left(\frac{2}{3}\right)^n$

Solution: Geometric series, $a_0 = 1$, $q = 2/3$. Sum $= \frac{1}{1-2/3} = 3$.

Exercise 1.6(): $\sum_{n=0}^{\infty} \left(\frac{7}{3^n} - \frac{6}{(n+3)(n+4)}\right)$

Solution: Geometric part: $\sum \frac{7}{3^n} = \frac{7}{1-1/3} = \frac{21}{2}$.

Telescoping part with $b_n = \frac{1}{n+3}$:

$$\sum_{n=0}^{\infty} \frac{6}{(n+3)(n+4)} = 6 \sum_{n=0}^{\infty} \left(\frac{1}{n+3} - \frac{1}{n+4}\right) = 6 \cdot \frac{1}{3} = 2.$$

Total: $\frac{21}{2} - 2 = \frac{17}{2}$.

Exercise 1.7: Investigate convergence of $\sum_{n=1}^{\infty} \sin(nx)$ for all $x \in \mathbb{R}$.

Solution (handwritten, partially legible): For $x = k\pi$, $k \in \mathbb{Z}$: $\sin(nx) = 0$ for all n. Converges trivially to 0.

For $x \neq k\pi$: Use the addition formula $\sin((n+1)x) = \sin(nx)\cos x + \sin x \cos(nx)$. Since $\sin x \neq 0$, we have $\lim_{n\to\infty} \sin(nx)\cos(nx) = 0$ because [**ILLEGIBLE: argument using** $\sin^2 + \cos^2 = 1$**; claim that** $\sin(nx)$ **does not converge but partial sums are bounded**].

The conclusion (typed at top of page): the series diverges for $x \neq k\pi$ [**ILLEGIBLE: full justification illegible; likely via necessary condition since** $\sin(nx) \not\to 0$**, or via partial sums argument**].

Exercise 1.6() [partially legible]: [**ILLEGIBLE: complex telescoping/geometric exercise involving** $(-\sqrt{2})^n$ **and**

n$^2 + n$; full solution not recoverable]

5.2 SIT2 — Tutorial Session 3, Sheet 2 (25.3)

The following are exercises from Sheet 2 solved in the 25.3 tutorial session.

Exercise 2.2(): $\displaystyle\sum_{n=2}^{\infty} \frac{(-1)^n}{\ln n}$

Solution: $b_n = 1/\ln n \searrow 0$. By Leibniz: converges. $\sum 1/\ln n > \sum 1/n$ (diverges by comparison), so not absolutely convergent. **Conditionally convergent.**

Exercise 2.2(): $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$

Solution: $b_n = 1/\sqrt{n} \searrow 0$. Leibniz: converges. $\sum 1/\sqrt{n}$ diverges (p-series, $p = 1/2$). **Conditionally convergent.**

Exercise 2.2(): $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}(n+1)}{n}$

Solution: $b_n = (n+1)/n \to 1 \neq 0$. Necessary condition fails. **Diverges.**

Exercise 2.2(): $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}n^3}{2^n}$

Solution: Ratio test on $|a_n| = n^3/2^n$:

$$\left|\frac{a_{n+1}}{a_n}\right| = \frac{(n+1)^3}{2n^3} \cdot \frac{1}{2} \cdot \frac{2^n}{1} = \frac{1}{2}\left(\frac{n+1}{n}\right)^3 \to \frac{1}{2} < 1.$$

Absolutely convergent.

Exercise 2.3(): $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n\sin(1/n)}$

Solution: $a_n = \frac{1}{n\sin(1/n)}$. Since $\sin(1/n) \sim 1/n$ as $n \to \infty$, we get $a_n \to 1 \neq 0$. Necessary condition fails. **Diverges.**

Exercise 2.3(): $\displaystyle\sum_{n=1}^{\infty} \frac{n+1}{n^2\sqrt{n+1}}$

Solution: $a_n \sim 1/n^{3/2}$. Limit comparison with $1/n^{3/2}$: $a_n \cdot n^{3/2} = n^{3/2}(n+1)/(n^2\sqrt{n+1}) \to 1$. Since $\sum 1/n^{3/2}$ converges ($p = 3/2 > 1$), the series **converges.**

Exercise 2.3(): [**ILLEGIBLE: series involving** n/e^n**; handwritten, illegible**]

Exercise 2.4(): $\sum_{n=1}^{\infty}(-1)^n \frac{(2n)!}{4^n (n!)^2}$

Solution (handwritten): Ratio test on $|a_n|$:

$$\left|\frac{a_{n+1}}{a_n}\right| = \frac{(2n+2)!}{4^{n+1}((n+1)!)^2} \cdot \frac{4^n (n!)^2}{(2n)!} = \frac{(2n+2)(2n+1)}{4(n+1)^2} = \frac{2n+1}{2(n+1)} \to 1.$$

Ratio test inconclusive. [**ILLEGIBLE: further analysis needed; likely shown not absolutely convergent by Stirling/Wallis, then Leibniz applied**]

6 Office Hours — Worked Proofs (23.3)

The following material was covered in the 23.3 office hour session.

6.1 Integral Test — Graphical Explanation (typed)

The key diagram: for f positive and decreasing with $f(n) = a_n$, the rectangle at $x \in [n, n+1]$ has area a_n. Comparing with the area under the curve:

$$\int_1^{\infty} f(x)\,dx \le \sum_{n=1}^{\infty} a_n \le a_1 + \int_1^{\infty} f(x)\,dx.$$

The series and integral converge/diverge together.

6.2 Limit Comparison Test — Proof Sketch (handwritten)

Given $b_n/a_n \to L$ with $0 < L < \infty$: for any $\varepsilon > 0$, eventually

$$(L-\varepsilon)\,a_n < b_n < (L+\varepsilon)\,a_n.$$

So $\sum a_n$ and $\sum b_n$ are bounded by multiples of each other. They behave identically.

6.3 Telescoping — General Formula (handwritten)

If $a_n = b_n - b_{n+1}$, then $S_n = b_1 - b_{n+1} \to b_1 - \lim b_n$.

Used in: $a_n = b_n - b_{n+1}$ where $b_n < a_n$; $\sum b_n$ and $\sum a_n$ controlled together when $b_n < a_n$ and $\sum b_n$ known.

6.4 Exercise 1.8(a) — Cauchy Criterion (handwritten)

Claim: $\sum_{n=1}^{\infty} \frac{\cos(nx) - \cos((n+1)x)}{n}$ converges for all x.

Solution: For $n > m$, the tail sum is:

$$\left|\sum_{k=m+1}^{n} a_k\right| = \left|\frac{\cos((m+1)x)}{m+1} - \frac{\cos((n+1)x)}{n} + \sum_{k=m+1}^{n} \cos((k+1)x)\left(\frac{1}{k+1} - \frac{1}{k}\right) + \cdots\right|$$

After telescoping and bounding $|\cos| \leq 1$:

$$\leq \frac{1}{m+1} + \frac{1}{n} + \frac{1}{m+1} - \frac{1}{n+1} \leq \frac{3}{N} < \varepsilon,$$

where $N = \lceil 3/\varepsilon \rceil$. Cauchy criterion holds for all x.

6.5 Exercise 1.9(b) — Divergence by Cauchy (handwritten)

Claim: $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n(n+1)}}$ diverges.

Solution: Take $\varepsilon = 1/5$, $m = N$, $n = 2N$:

$$S_{2N} - S_N = \sum_{k=N+1}^{2N} \frac{1}{\sqrt{k(k+1)}} \geq \frac{N}{\sqrt{2N \cdot (2N+1)}} > \frac{N}{2N+1} > \frac{1}{5}.$$

Cauchy criterion fails. Diverges.

[**ILLEGIBLE: Additional office hour content — several pages of handwritten derivations not fully legible; appeared to involve further convergence test examples**]

7 Quick Reference: Convergence Test Decision Tree

1. **Necessary condition**: $a_n \not\to 0 \Rightarrow$ diverges immediately.
2. **Positive series**: try comparison, limit comparison, ratio, root, integral.
3. **Alternating series**: check Leibniz ($b_n \searrow 0$) for conditional convergence; apply ratio/root to $|a_n|$ for absolute convergence.
4. **General series**: apply ratio/root to $|a_n|$. If $L = 1$, use Dirichlet or Abel.
5. **When ratio test fails** (oscillating ratio): use generalized root test ($\limsup$).
6. **Telescoping**: recognize $a_n = b_n - b_{n+1}$; sum $= b_1 - \lim b_n$.
7. **Geometric**: $\sum aq^n$ converges iff $|q| < 1$; sum $= a/(1-q)$.